Time Translations

In class we learned that the (linear) momentum p is the "generator" of translations in space. (We work in one spacial dimension for simplicity.) Concretely what this meant is that we can construct from p an operator U on the Hilbert space of functions $\{f(x)\}$ of the coordinate x so that

$$U(a)f(x) = f(x+a)$$

Recall that since we want to preserve the Hilbert space norm, U should be unitary. From this and certain limits, we worked out that $U(a)$ could be expressed in terms of the absolutely convergent sum for the exponential

$$\exp\left(\frac{i}{\hbar}ap\right) = \sum_{n=0}^{\infty}\frac{1}{n!}\left(\frac{i}{\hbar}ap\right)^n$$

In other words $U(a) = e^{a\frac{d}{dx}}f(x) = f(x+a)$.

0. Show this explicity by expanding out the exponential and reinterpreting the result as a Taylor series.

The exponential sum is

$$e^{a\frac{d}{dx}} = \sum_{n=0}^{\infty}\frac{1}{n!}a^n\frac{d^n}{dx^n}$$

Let $f(x)$ be any smooth function and Taylor expand $f(x+a)$ around some constant a:

$$f(x+a) = \sum_{n=0}^{\infty}\frac{1}{n!}a^n f^{(n)}$$

where $f^{(n)}$ is a common notation for the nth derivative of f. But then we are done, because these two expansions are identical.

In this set of notes, we will work out the same set of steps for translations in time. That is, we aim to find the generator of time translations. Time in Newtonian physics is nothing like space. All observers agree on the passing of time, and at any particular moment in time, space is Euclidean. Coordinates like x are merely choices we make to parameterize this space. Time on the other hand defines *dynamics* though Newton's Second Law. Because of this, what we will find that the rabbit hole goes far deeper in time...

So, by complete analogy, we are looking for an operator $U(c)$ that maps functions of time $f(t) \mapsto f(t+c)$ for any constant c. By all the same arguments, this has to have the form $U(c) = e^{c\frac{d}{dt}}$ which we can write as $e^{-\frac{i}{\hbar}ch}$ where h is a Hermitian operation on functions called the generator of time translations. Just as p was the generator of translations, we now need to identify this operator.

1. Find the dimensions of the generator h. What real (*i.e.* has no imaginary part) quantity do you know with these dimensions?

Let $[A]$ denote "the dimensions of A". Exponentiation of a dimensionful quantity is meaningless (ask, if you don't understand this!), so $[\frac{i}{\hbar}ch] = 1$. The imaginary unit is dimensionless since it is just a number. $[\hbar] = [x] \cdot [p] = L \cdot ML/T = ML^2/T$. Since c enters in $f(t) \mapsto f(t+c)$, $[c] = [t] = T$. Thus $1 = [\frac{ch}{\hbar}] = T/(ML^2) \cdot T \cdot [h]$, which implies $[h] = ML^2/T^2$. But these are the dimensions of mass times velocity-squared, which is an energy. So whatever h is, it must have the dimesions of energy.

Now to identify what h actually is, we use the observation that the generator of time translations is what we use to write Newton's second law

$$F = \frac{dp}{dt}$$

and it is unique in that way. We must also recall that the momentum is *defined* as a derivative of the kinetic energy K.

2. If you did not derive this in class, derive explicitly from the example of a free classical particle of mass m and velocity $\mathbf{v}$ a formula for $\mathbf{p}$ as a derivative of K.

The energy of a free particle is $K = \frac{1}{2}mv^2$ where $v = \sqrt{\mathbf{v} \cdot \mathbf{v}}$. Thus, component-by-component writing $\mathbf{v} = (v^i)$ for the 3 differnt directions $i = x, y, z$, $\frac{\partial K}{\partial v^i} = 2 \times \frac{1}{2}mv_i = mv_i = p_i$. Thus, component-by-component

$$\mathbf{p} = \frac{\partial K}{\partial \mathbf{v}}$$

We can take this formula as the definition of $\mathbf{p}$. We say that "$\mathbf{p}$ is the momentum canonically conjugate to $\mathbf{x}$". (More on this terminology below.)

Besides the kinetic energy there is another potential energy for any conservative force: If $\mathbf{R}^3$, $0 = \int_\gamma \mathbf{F} \cdot d\mathbf{x} = 0$ for any closed contour γ, then there is a function $U(\mathbf{x})$ such that $\mathbf{F} = -\nabla U$. Importantly, although $K = K(x, v)$ is allowed to depend on x, $U(x)$ is *not* allowed to depend on v. (I know, unfair right?) Now consider the total energy function $H = K(x, v) + U(x)$.

3. For this problem, treat p and v as independent variables.

a. Show that for any kinetic energy, the combination

$$H(x, p) = pv - K(x, v) + U(x)$$

is independent of v.

b. Assume that $K(x, v) = f(x)v^\alpha$ for some real number α. Show that $pv - K = (\alpha - 1)K$.

c. Combine parts a. and b. to conclude that, if K is quadratic in v, then $H(x, p) = K(x, v) + U(x)$.

3 a. Just calculate the partial derivative and use the definition of the momentum:

$$\frac{\partial H}{\partial v} = \frac{\partial}{\partial v}\left[pv - K(x, v) + U(x)\right] = p - \frac{\partial H}{\partial v} = p - p = 0$$

b. If $K = f(x)v^\alpha$, then $\frac{\partial K}{\partial v} = \alpha f(x)v^{\alpha-1}$ which implies that $v\frac{\partial K}{\partial v} = \alpha K$. But this is the same as $pv = \alpha K$, so $pv - K = (\alpha - 1)K$.

c. If K is quadratic in v, then $\alpha = 2$. When that happens, $pv - K = (\alpha - 1)K = K$, so $H(x, p) = pv - K(x, v) + U(x) = K(x, v) + U(x)$.

This took a lot of effort that seems like it should not have been necessary, so we should go back an look at how K was originally defined.

Recall that all these energy consideratoins came from work without dissipative forces. Then $F = -\nabla U$ for some U and $\int_\gamma F \cdot dx = -\int_\gamma \nabla U \cdot dx = U|_{-\partial\gamma}$. But $\int_\gamma F \cdot dx = \int_\gamma \dot{p} \cdot dx = \int_\gamma m\dot{v} \cdot vdt = \frac{m}{2}\int_\gamma \frac{dv^2}{dt}dt = \frac{m}{2}v^2\big|_{\partial\gamma}$. Thus $\left[\frac{m}{2}v^2 + U\right]_{\partial\gamma} = 0$ and this holds along any path γ in space. Therefore, $\frac{m}{2}v^2 + U = \text{constant}$. This was the original motivation for introducing $K = \frac{m}{2}v^2$.

Now in general

$$\begin{aligned}\int_\gamma F \cdot dx &= \int_\gamma \dot{p} \cdot dx = \int_\gamma \frac{d}{dt}\left(\frac{\partial K}{\partial v}\right) \cdot dx = \int_\gamma \frac{d}{dt}\left(\frac{\partial K}{\partial v}\right) \cdot vdt \\ &= \int_\gamma \frac{d}{dt}\left(\alpha f(x)v^{\alpha-1}\right) \cdot vdt = \int_\gamma \frac{d}{dt}\left(f(x)v^\alpha\right)dt = \int_\gamma \frac{dK}{dt}dt = K|_\gamma \end{aligned} \tag{1}$$

so we, again recover the work-kinetic energy theorem, but this time for any kinetic energy of the form $K = f(x)v^\alpha$.

Hamiltonian formulation of dynamics

But wait, what?! H clearly depends on v! What's going on??

4. Rewrite the kinetic energy part of the Hamiltonian H for a particle of mass m in terms of momentum. Use this expression to show that

$$\frac{dx}{dt} = \frac{\partial H}{\partial p}$$

Speculate about what this means for the dependence of the total energy on position, velocity, and linear momentum.

The kinetic energy of a particle of mass m is $K = \frac{m}{2}v^2$. Its momentum is is $p = \partial K/\partial v = mv$, to $v = \frac{p}{m}$, and $K = \frac{1}{2}pv = \frac{1}{2m}p^2$. Thus $dx/dt = v = \partial K/\partial p$, but since the potential energy does not depend on v, it cannot depend on p. Therefore, $dx/dt = \partial H/\partial p$.

6. For a partcile of mass m in Euclidean space, show that Newton II can be re-expressed as

$$\frac{dp}{dt} = -\frac{\partial H}{\partial x}$$

This is again just a calculation:

$$\frac{\partial H}{\partial x} = \frac{\partial K}{\partial x} + \frac{\partial U}{\partial x} = -F = -\frac{dp}{dt}$$

Congratulations! You have derived Hamiltons "canonical equations". Let's recap. The total energy function $H(x,p)$ is a function on the **phase space** of positions and momenta. This space is $2n$-dimensional when the **configuration space** (a.k.a. "space", *e.g.* $\mathbf{R}^3$) is n-dimensional. In this formulation of Newtonian physics, particle dynamics corresponds to the allowed paths in phase space. If there is no *explicit* time-dependence in the system (*i.e.* the system is not being driven with time-dependent forces), then all intersting dynamical quantities can be formulated as functions on this phase space. So all the quantities we care about are phase space functions. (We will return to explicit time-dependence later.)

In this phase space formulation, the dynamics of any quantity $f(x,p)$ is encoded in Hamilton's equations. That is, the "Hamiltonian (function)" completely determines the dynamics of the system.

7. Let $f(x,p)$ be any differentiable function on the phase space. Use the chain rule and Hamilton's equations to write a formula for df/dt.

$$\frac{df}{dt} = \frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial p}\frac{dp}{dt} = \frac{\partial f}{\partial x}\frac{\partial H}{\partial p} - \frac{\partial f}{\partial p}\frac{\partial H}{\partial x} \tag{2}$$

Poisson Bracket

Looking at this formula we found in problem 6, we notice an interesting structure that encourages us to introduce the **Poisson bracket**: For any two phase space functions f and g, define the function

$$\{f,g\} = \frac{\partial f}{\partial p}\frac{\partial g}{\partial x} - \frac{\partial f}{\partial x}\frac{\partial g}{\partial p}$$

8. Rewrite your dynamical equation in terms of the Poisson bracket.

Almost by inspection, the equation can be written as

$$\frac{df}{dt} = \{H, f\}$$

It is important to note that we did not introduce any new structure into the theory "by hand". (An example of a structure that is just thrown in by hand is the cross product in particle mechanics.) Even the dot product is not a native thing to the vector space $\mathbf{R}^3$, but added as an assumtion that makes the vector space into an "inner product space" we can Euclidean. The Poisson bracket, by contrast, is a stucture that comes with any phase space *automatically*. It was handed to us by the math Gods as a gift. This is the reason mathematicians call it "canonical", and this particular type of canonical structure is called "symplectic" in the math jargon.

9. For any 3 phase space functions f, g, and h, check that the Poisson bracket satisfies the following properties:

i. $\{g, f+h\} = \{f,g\} + \{f,h\}$ and for any real number $c \in \mathbf{R}$, $\{f, cg\} = c\{f,g\}$

ii. $\{g,f\} = -\{f,g\}$

iii. $\{f,\{g,h\}\} + \{g,\{h,f\}\} + \{h,\{f,g\}\} = 0$

The first property just follows from how derivatives behave. The second property is obvious. The third is an annoying calculation:

$$\begin{aligned}\{h,\{f,g\}\} &= \left\{h, \frac{\partial f}{\partial p}\frac{\partial g}{\partial x} - \frac{\partial f}{\partial x}\frac{\partial g}{\partial p}\right\} \\ &= \left\{h, \frac{\partial f}{\partial p}\right\}\frac{\partial g}{\partial x} - \left\{h, \frac{\partial f}{\partial x}\right\}\frac{\partial g}{\partial p} + \frac{\partial f}{\partial p}\left\{h, \frac{\partial g}{\partial x}\right\} - \frac{\partial f}{\partial x}\left\{h, \frac{\partial g}{\partial p}\right\} \end{aligned} \tag{3}$$

Since this is such a mess, we simplify the notation by writing the derivatives as subscripts:

$$\begin{aligned}
\{h,\{f,g\}\} &= \left\{h,f_p\right\}g_x - \{h,f_x\}\,g_p + f_p\,\{h,g_x\} - f_x\left\{h,g_p\right\} \\
&= h_p f_{xp} g_x - h_x f_{pp} g_x - h_p f_{xx} g_p + h_x f_{px} g_p \\
&+ f_p h_p g_{xx} - f_p h_x g_{px} - f_x h_p g_{xp} + f_x h_x g_{pp} \\
&= f_{xx}\left(-g_p h_p\right) + f_{xp}\left(g_p h_x + g_x h_p\right) + f_{pp}\left(-g_x h_x\right) \\
&+ g_{xx}\left(h_p f_p\right) + g_{xp}\left(-h_x f_p - h_p f_x\right) + g_{pp}\left(h_x f_x\right)
\end{aligned} \tag{4}$$

Now we need to add the cyclic permutations $f \to g \to h \to f$, but here is what will happen: Note that if we do this permutation to the first line of the result, we get the second line with all the signs switched. This means that when we add up the permutations, the second line gets canceled by the first line of the next permutation, and this repeats, so everything cancels when we add up all three.

If that was too slick for you, I'll do it explicitly:

$$\begin{aligned}
& f_{xx}\left(-g_p h_p\right) + f_{xp}\left(g_p h_x + g_x h_p\right) + f_{pp}\left(-g_x h_x\right) \\
&+ g_{xx}\left(h_p f_p\right) + g_{xp}\left(-h_x f_p - h_p f_x\right) + g_{pp}\left(h_x f_x\right) \\
\\
&+ g_{xx}\left(-h_p f_p\right) + g_{xp}\left(h_p f_x + h_x f_p\right) + g_{pp}\left(-h_x f_x\right) \\
&+ h_{xx}\left(f_p g_p\right) + h_{xp}\left(-f_x g_p - f_p g_x\right) + h_{pp}\left(f_x g_x\right) \\
\\
&+ h_{xx}\left(-f_p g_p\right) + h_{xp}\left(f_p g_x + f_x g_p\right) + h_{pp}\left(-f_x g_x\right) \\
&+ f_{xx}\left(g_p h_p\right) + f_{xp}\left(-g_x h_p - g_p h_x\right) + f_{pp}\left(g_x h_x\right)
\end{aligned} \tag{5}$$

Now we finally see that all the terms enter twice with opposite signs, so they cancel pairwise.

Recapitulation

The fundamental equations of Newtonian dynamics can be reformulated as statements about operations on a linear space $\mathscr{H}$ that almost magically comes with a Poisson bracket $\{\cdot,\cdot\}$, that promotes it to a very special sort of algebra that is intimately tied to a specific class of differential equations and their solutions. In this formalism, time translations are generated by a distinguished function called the Hamiltonian.

This space is not just a vector space with an norm (*i.e.* the Hilbert space part), but a Lie algebra because it automatically comes with a Poisson bracket that satisfies the defining relations of a Lie bracket.

Time dependence

Hamilton's formulation of dynamics shows that the time evolution of any function $f \in \mathcal{H}$, that is any function f on the phase space of ps and xs is generated, through the Poisson Bracket, by the Hamiltonian. This, however, is not the most general behavior because it is always possible to have explicit dependence on time. To describe this situation, we talk about the *extended* phase space of xs, ps, and t.

10. a. Show that for a function on the extended phase space

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \{f, H\}$$

  b. Use this to show that, unless the Hamiltonian H depends explicitly on time, it is **conserved**, where "conserved" means constant in time.

9 a. A function on the extended phase space has the form $f(x, p, t)$. Therefore, $\frac{df}{dt} = \frac{\partial f}{\partial t} +$ parts already there when it does not depend explicitly on t. But we already know these are generated by H through the Poisson bracket. This gives the stated equation.
b. Plugging in $f \to H$ gives

$$\frac{dH}{dt} = \frac{\partial H}{\partial t} + \{H, H\} = \frac{\partial H}{\partial t}$$

The $\{H, H\} = -\{H, H\}$ term vanishes by antisymmetry of P.B., so $\frac{dH}{dt} = 0$ iff $\frac{\partial H}{\partial t} = 0$.

Hamiltonian quantization

11. Suppose we have an algebra of operators $\mathcal{O}$ with a multiplication that may or may not be associative: For any f, g, $h \in \mathcal{O}$

$$(h(fg)) \stackrel{?}{=} ((hf)g)$$

and define the **commutator**

$$[f, g] = fg - gf$$

Show that $[\cdot, \cdot]$ satisfies the Jabobi identity if the algebra multiplication is associative.

Let's look at the first of three terms

$$[h, [f, g]] = (h(fg)) - ((fg)h) + ((gf)h) - (h(gh))$$

The other two terms are the permulations $f \to g \to h \to f$ and $f \to h \to g \to f$.

An example of an associative algebra is the algebra of matrices with matrix multiplication.